

AXEL BERRAL LOPEZ

AI ENGINEER | DATA SCIENCE

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PROFESSIONAL PROFILE

AI Engineer experienced in designing, developing, and deploying end-to-end AI systems, **local LLMs**, RAG, and speech/NLP pipelines in on-premise environments. Background in backend services, APIs, and GPU inference optimization, with a focus on performance, scalability, and production systems.

PROFESSIONAL EXPERIENCE

- AI Engineer** | MST HOLDINGFeb 2026 - Present
- Designed and developed AI-based call analysis systems for automated agent evaluation, processing thousands of recordings.
 - Built end-to-end audio pipelines: ingestion, Whisper transcription, routing/NLP, and LLM-based evaluation.
 - Integrated and deployed local LLMs on on-premise infrastructure, with APIs and backend services for orchestration.
 - Optimized GPU inference performance and scalability, deploying and maintaining production AI pipelines.
- AI Engineer Intern** | StiketsJun 2025 - Aug 2025
- Built a modular framework for LLM-based AI agents.
 - Improved backend and frontend functionality to integrate the framework.

EDUCATION

- Master's Degree in Data Science** | Universitat Oberta de Catalunya (UOC)Sep 2026 - Jun 2027
- BSc in Computer Engineering** | Universitat de Lleida (UdL)2022 - 2026
- Erasmus Exchange** | NTNU, Gjøvik (Norway) | 2025

SELECTED PROJECTS

- BSc Thesis - Thermal Prediction with MLP, GNN and Tiny-GNN+PINN.** 2D heat-diffusion modeling for edge computing; Full-GNN reduced MSE by ~40% vs MLP, while Tiny-GNN+PINN improved Tiny-GNN by 41.7% at 20% noise.
- LoveLink.** Recommendation-based social network using machine-learning algorithms and friend-of-a-friend relationships.

TECHNICAL SKILLS

- AI / ML:** LLMs, RAG, LangChain, Ollama, vLLM, TensorRT-LLM, Whisper, Scikit-learn, Pandas
- Backend / Data:** Python, FastAPI, Flask, SQL, PostgreSQL, MongoDB, Neo4j, Firebase
- Engineering / Infra:** Docker, Git, Celery, RabbitMQ, REST APIs, Linux, Java, JavaScript

LANGUAGES

- Spanish** (Native) | **Catalan** (Native) | **English** (B2) | **Norwegian** (A1)